

what are some of the key columns to look for when doing analytics for health institutions

When doing analytics for a healthcare institution (hospital, clinic, medical practice, or healthcare group), the most valuable columns depend on the business problem you're trying to solve. However, there are several core categories of data that are commonly used for operational, financial, clinical, and patient analytics.

1. Patient Demographics

These help understand patient populations and segment patients.

Key Columns:

- Patient_ID
- Age
- Gender
- Race/Ethnicity (where legally permitted)
- Marital Status
- Occupation
- Residential Area/Suburb
- Province/Region
- Medical Aid/Insurance Type

Analytics Use Cases:

- Patient segmentation
- Population health analysis
- Disease prevalence by age group
- Geographic service demand



+ Message Copilot





Patient-Centric Data Model

The **Patient_ID** is usually the primary key that links most healthcare datasets together.

Patient Table (Parent Table)

Patient_ID	Name	DOB	Gender	Medical_Scheme
P1001	John Smith	1980-03-15	M	Discovery
P1002	Sarah Jones	1992-07-01	F	Bonitas

Primary Key

Patient_ID

Every patient should have one unique Patient_ID.

Admissions Table

A patient can have many admissions.

Admission_ID	Patient_ID	Admission_Date	Discharge_Date
A1001	P1001	2026-01-15	2026-01-20
A1002	P1001	2026-05-10	2026-05-12

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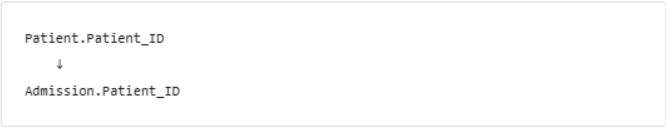


Admissions Table

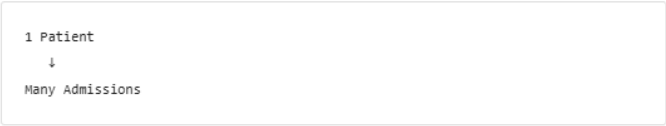
A patient can have many admissions.

Admission_ID	Patient_ID	Admission_Date	Discharge_Date
A1001	P1001	2026-01-15	2026-01-20
A1002	P1001	2026-05-10	2026-05-12

Relationship



One-to-Many relationship:

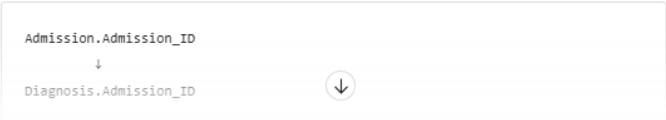


Diagnosis Table

Each admission can have multiple diagnoses.

Diagnosis_ID	Admission_ID	ICD10_Code
D001	A1001	E11.9
D002	A1001	I10

Relationship



detail how i primary keys and foreign keys are used in this scenario

In a healthcare analytics environment, **Primary Keys (PKs)** and **Foreign Keys (FKs)** are what allow you to connect data from different systems into a single patient journey.

Think of it this way:

- A **Primary Key** uniquely identifies a record in a table.
- A **Foreign Key** is a field that references a Primary Key in another table.

Without these keys, you cannot reliably join admissions, diagnoses, billing, lab results, and outcomes together.

1. Patient Table (Parent Table)

This is usually the master record.

Patient

Patient_ID (PK)	Name	DOB	Gender
P1001	John Smith	1980-04-10	M
P1002	Sarah Jones	1992-07-18	F

Primary Key



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1 Patient_ID
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Why?

Every patient must be uniquely identifiable.

Example:



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1 P1001 = John Smith
2 P1002 = Sarah Jones
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2. Admission Table

A patient may be admitted many times.

Admission

Admission_ID (PK)	Patient_ID (FK)	Admission_Date
A1001	P1001	2026-01-15
A1002	P1001	2026-04-20
A1003	P1002	2026-05-10

Primary Key



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1 Admission_ID

Foreign Key



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1 Patient_ID

This creates:



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1 Patient
2 ↓
3 Admission

Relationship:



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1 Patient_ID (PK)
2 ↓
3 Patient_ID (FK)



Meaning:

+ +

